

Kian Javaheri

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EDUCATION

Arizona State University

Aug. 2022 – May 2026

B.S. in Computer Science, Economics

Tempe, AZ

- **GPA: 3.93** · Summa Cum Laude | Barrett, The Honors College

EXPERIENCE

Sandia National Laboratories

May 2023 – Aug. 2024

R&D Bioresource and Environment Security Intern

Livermore, CA

- Led and designed a standalone **optimization tool in Python** that ingests operational data across multiple bioresource systems and maximizes projected cash flow via configurable optimization algorithms
- Created and validated shapefile datasets, then engineered a companion **QGIS plugin** in Python that connects to the optimization tool via an internal API, letting users gauge how different feedstock locations affect profit yields

Dr. Michael Hanemann's Team @ Arizona State University

May 2025 – May 2026

Junior Researcher

Tempe, AZ

- Automated large-scale document retrieval from the Arizona Department of Water Resources using **Selenium WebDriver**, eliminating manual collection across hundreds of annual reports
- Engineered **OCR extraction** pipelines to parse and structure tabular data from scanned PDFs, producing clean, analysis-ready datasets for downstream econometric modeling

SELECTED PROJECTS

Revolution Parts Shipping Service with Caching | *React, PostgreSQL, Flask, Docker*

- Architected a high-throughput relational caching layer using **PostgreSQL** and **Flask** within a **Dockerized** microservice environment, intercepting and caching external carrier API responses to reduce redundant network calls
- Achieved a **64.4% reduction in processing latency** (modal response time: 334ms → 119ms), delivering a **2.8x speedup** over live API calls and substantially reducing data-layer overhead

Materials GUI | *Java, Fabric, Claude API*

- Developed a client-side **Fabric mod** for Minecraft that tracks the materials needed for a build, letting players import a material list by pasting text, loading a Litematica export, or dropping in a screenshot parsed via **OCR** through the **Claude API**
- Implemented real-time inventory tracking across in-game storage containers, color-coding each slot by completion progress and auto-routing shift-clicked items to the slots still needing materials

LEADERSHIP & ACTIVITIES

Arizona State University

Aug. 2023 – May 2024

Undergraduate Teaching Assistant

Tempe, AZ

CSE310: Data Structures & Algorithms

- Led exam review sessions on **red-black trees**, **dynamic programming**, and **minimum spanning tree** algorithms, and held 5+ office hours per week guiding students through algorithm design challenges

FSE150: Intro to Grand Challenges

- Facilitated weekly engineering discussions for a class of ~30, reinforcing interdisciplinary thinking within an engineering context

Google Data Analytics Certificate

Oct. 2022

Coursera

- Completed all 8 courses including data cleaning and **SQL querying in R**

TECHNICAL SKILLS

Machine Learning & AI: Artificial Intelligence, Fine-Tuning, Transformers, PyTorch, PySpark, Predictive Modeling

Languages: Java, Python, C/C++, SQL, JavaScript/TypeScript, HTML/CSS, R

Frameworks: React, Node.js, Flask, JUnit

Developer Tools: Git, GitHub, Docker, JupyterHub, Firebase, MATLAB, QGIS, Postman, Stata, Qualtrics

Libraries: Pandas, NumPy, Matplotlib, Selenium